HridayMitra check-in stool: system block diagram

Home early-warning monitor for worsening heart failure (monitoring, not diagnosis). Custom part = 8-electrode bioimpedance analog front end (AFE) + ESP32.

PATIENT SIDE: applied part, IEC 60601-1 type BF

battery only while touched · SW6 isolates the charger

Electrodes J1 ×8 — SS304 plates

4 foot plates: forefoot = current,

heel = sense (tetrapolar)

4 hand strips: outer = current,

inner = sense

skin contact 0.2-10 kΩ tolerated

4 paths: legs, arms, R side, L side

Finger optical — MAX30102

red 660 nm + IR 880 nm LEDs

18-bit, 100 S/s, 7 mA LED

Seat load cells ×4 — 50 kg half-bridges

one full bridge → HX711

24-bit, 10 S/s, 105 counts/kg

User interface

GREEN / RED answer + SOS buttons

tablet-drawer microswitch

4 current

4 sense

CUSTOM AFE BOARD (this design, HM-HW-002)

Path switches — 74HC4052 U4/U5

U4 current pair

U5 sense pair

Ron ≈ 100 Ω (cancels)

Current source — MCP6002 U1

DAC → 1 µF → buffer

C3 + C4 DC block

R4 4.7 k: 100 µA rms

Body / ref select — 74HC4053 U3

A/B: body or reference

C: Rcal or R5+Rcal

(self-check ratio 2.2)

Instrumentation amp — INA126 U2

G = 10 (RG 16 k)

R8/R9 10 MΩ bias

1.5 k / 10 n anti-alias

References

Rcal 100 Ω 0.1 % 25 ppm: the impedance reference

R5 120 Ω: self-check partner (ratio 2.2 ± 9 %)

VREF 1.65 V: R1/R2 10 k 1 % + U1A buffer

Z = Rcal · |V(body)| / |V(Rcal)|

(current, gain and Ron cancel)

DAC1 cosine

ADC 1.65±0.6 V

S0 S1 SEL CAL

I²C 400 kHz · MAX30102 @ 0x57

HX711 2-wire, 24-bit @ 10 S/s

GPIO 4 / 5 / 15 / 36

ESP32-WROOM-32 (DevKitC)

Xtensa LX6 ×2, 240 MHz · 520 KB SRAM · 4 MB flash

DAC1 GPIO25: 5/20/50 kHz cosine

ADC1_CH7 GPIO35: DMA 200 kS/s

GPIO33/27 S0 S1 · 18 SEL · 12 CAL

I²C GPIO21/22 · HX711 GPIO32/14

UART2 GPIO16/17 · DAC2 GPIO26 voice

GPIO39 charge sense · 36 drawer

Firmware (C++, offline):

· Goertzel + Cole fit (EWI)

· alert engine v3 (= Python, 100 %)

· care-loop state machine, SMS queue

· self-test, self-calibration

· BLE: amulet scan + app GATT

· flash: 2 users' history, 25 voice clips

((·)) BLE GATT → Saathi phone app (6e8a0001-…)

((·)) GSM 900/1800 → family, ASHA, PHC doctor

UART2 9600 8N1

GSM modem — SIM800L

2G quad-band SMS · Airtel / Vi / BSNL

2 A bursts from VBAT, 1000 µF

RTC from network (AT+CCLK)

DAC2

Voice out — PAM8403 + 4 Ω 3 W

8-bit 8 kHz clips from flash

Hindi / English, per user

GPIO

Status — LEDs + ERM motor

green / amber / red, 220 Ω

seat vibration via 2N2222

USB 2.0 FS

USB-UART (on DevKit) — CP2102

laptop bridge, flashing, app (USB)

laptop on battery only

BLE 2.4 GHz adv

Kavach amulet — ESP32-C3 + MPU6050

breathing, sit-ups, 6-min walk, fall, SOS

POWER TREE

USB-C in — 5 V

phone charger

5 V

SW6 CHARGE/USE — DPDT

USE: both wires cut

CHARGE: GPIO39 high

Li-ion charger — TP4056

1 A CC/CV, 4.2 V

DW01 protection

18650 cell — 2500 mAh

3.0-4.2 V (VBAT)

SW4 power switch

VBAT

Boost — MT3608

VBAT → 5.0 V

≈ 90 % efficient

5 V

LDO (DevKit) — AMS1117-3.3

5 V → 3.3 V, 1 A

AFE + sensors

3V3

VREF — U1A buffer

3V3 / 2 = 1.65 V

AFE mid-rail

Rails: VBAT → SIM800L direct (2 A bursts) · 5 V → DevKit VIN, PAM8403, motor · 3V3 → AFE, MAX30102, HX711 · VREF → AFE bias

Budget (HM-HW-003, datasheet typical): idle 130 mA → 0.7 days per charge as built; Rev B power plan 22 mA → 3.9 days · check-in 345 mA · peak 2.3 A (GSM)

LEGEND

- VBAT 3.0-4.2 V
- 5 V
- 3V3
- VREF 1.65 V
- signal / bus
- radio link

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Doc HM-HW-001	Rev A 2026-10-25
Sheet 1 / 1	Transfinitte 2025, hardware